BIOL 230: Pacific Spirit Park Lab Report

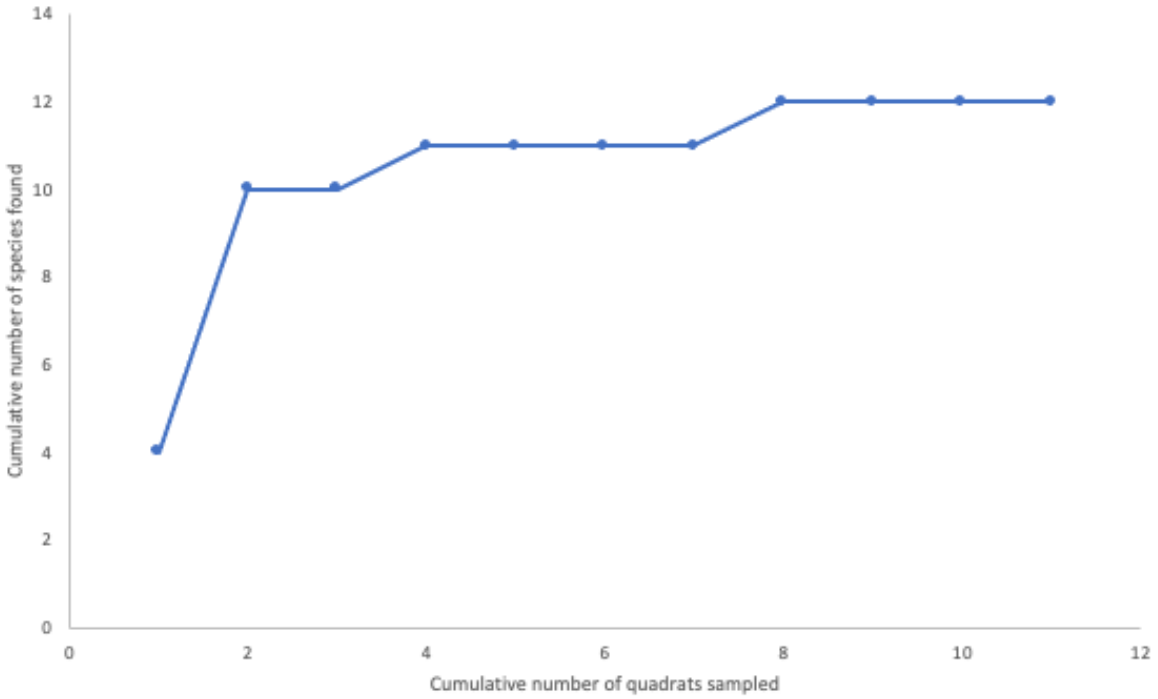


Figure 1. The X-axis shows the cumulative number of 1 m × 1 m quadrats sampled (from the 20 m position toward the trail), and the Y-axis shows the cumulative number of unique understory plant species recorded, including the "other" (non-target) category. Each point represents the total number of distinct species observed after that number of quadrats was surveyed. The curve levels off at 12 species by the 8th quadrat, indicating that few or no new species were encountered afterward. Sample size = 11 quadrats. Data were collected on 29 September 2025 in Pacific Spirit Park, Vancouver, BC.

Table 1. Average number of pollution observations (dog feces and garbage) recorded in west- and east-side transects across all class data collected at Pacific Spirit Park on 29 September 2025. Columns indicate mean counts of dog feces and garbage observed per transect, rounded to two decimal places. West transects are located closer to the forest edge, whereas east transects are farther inside the forest.

Direction	Dog Feces Avg	Garbage Avg
W	0.62	1.48
E	0.19	0.71

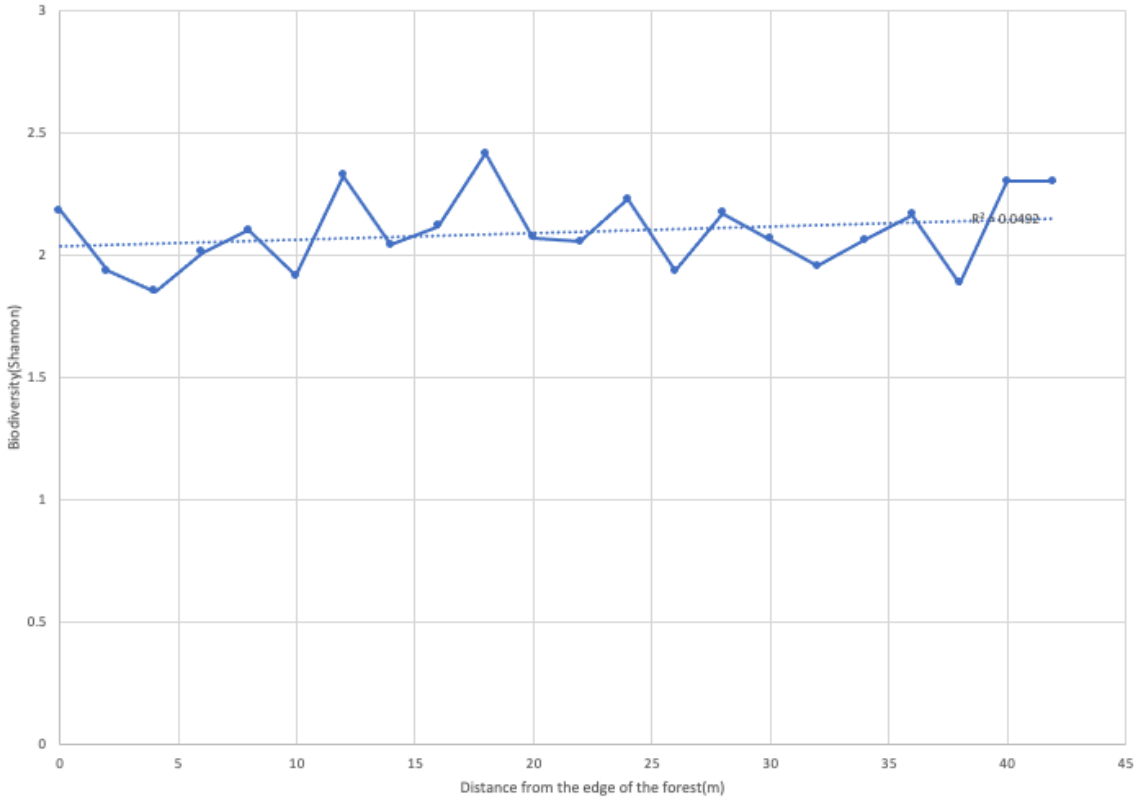


Figure 2. Biodiversity (Shannon index) plotted against distance from the forest edge at Pacific Spirit Park. Each point represents the Shannon diversity index (H) calculated from the average percent cover of understory plant species at a given distance (0–42 m). The dotted line shows the linear trendline ($R^2 = 0.0492$). Shannon index at 42 m = 2.30. Sample size = 22 quadrat locations. Data were collected on 29 September 2025 in Pacific Spirit Park, Vancouver, BC.

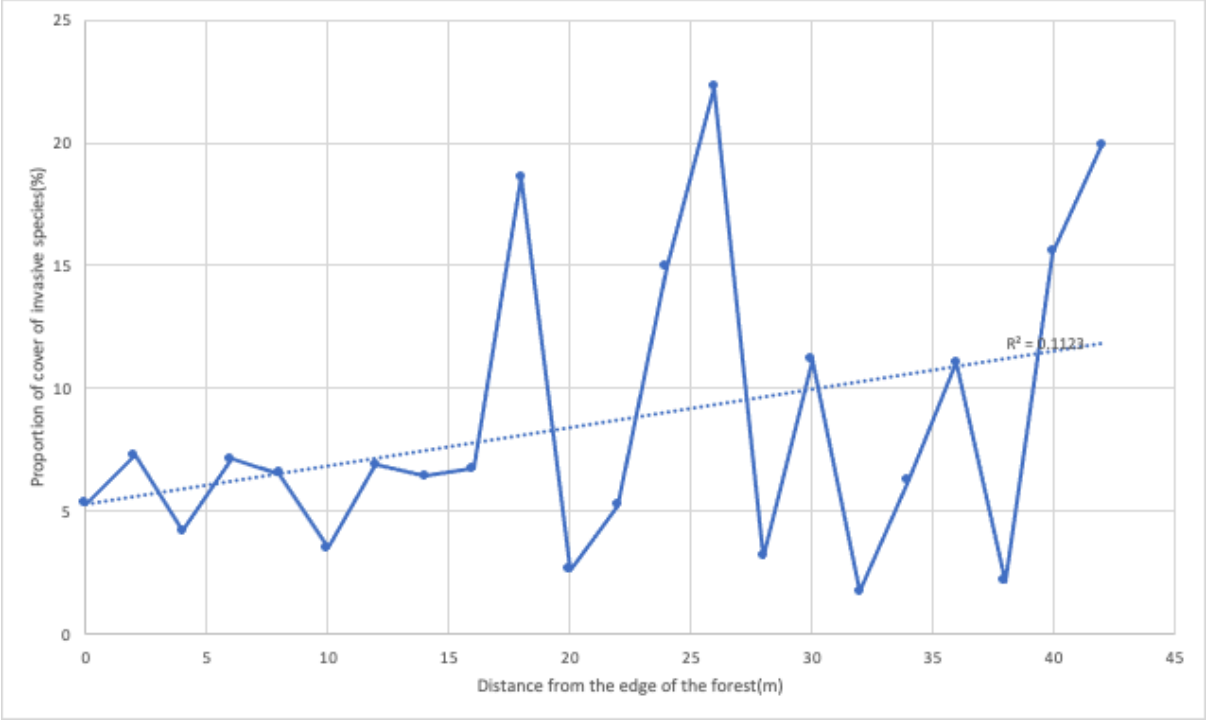


Figure 3. Proportion of total plant cover made up of invasive species plotted against distance from the forest edge at Pacific Spirit Park. Each point represents the percentage of total understory cover composed of invasive species (English holly, Himalayan blackberry, English laurel, and wall lettuce) at a given distance (0–42 m). The dotted line indicates the linear trendline ($R^2 = 0.1123$). Sample size = 22 quadrat locations. Data were collected on 29 September 2025 in Pacific Spirit Park, Vancouver, BC.

Questions:

1. The species along our transect were adequately sampled. In our Figure 1, the species-accumulation curve rose sharply at first and then plateaued around the 8th quadrat, stabilizing at 12 species. This leveling-off pattern indicates that additional sampling yielded few or no new species, suggesting that most of the understory species present in the transect were captured. By contrast, in the reference plot on page 4, inadequate sampling is shown by a curve that continues to rise without reaching an asymptote. Because our curve reached a clear asymptote, it demonstrates that our sampling effort was sufficient to represent the community's species richness.

2. Pollution was higher at the forest edge. West-side transects, which are closer to the forest boundary, had higher average counts of both dog feces (0.62) and garbage (1.48) than east-side transects farther inside the forest (0.19 and 0.71, respectively). This pattern matches expectations, as areas near trails and residential edges experience greater human and pet activity, leading to more waste and disturbance. During field work, we also observed more visible trampling and litter near the western edge, supporting the pattern seen in Table 1.

3. Hypothesis 1 – Biodiversity increases with distance from the forest edge. Disturbance, pollution, and trampling are strongest at the edge, favouring only a few stress-tolerant or invasive r-selected species. Deeper in the forest, environmental conditions become more stable and shaded, allowing more K-selected, shade-tolerant species to coexist. Thus, species diversity (Shannon H) should rise with distance from the edge.
Hypothesis 2 – Biodiversity decreases with distance from the forest edge. The edge may host a mixture of native and opportunistic species, producing higher richness. Farther inside, stronger competition among a few dominant shade-tolerant species could reduce overall diversity.
Interpretation of Figure 2 – The trendline shows a weak positive relationship between distance from the forest edge and Shannon diversity ($R^2 = 0.0492$, $r \approx +0.22$). This means that only about 5 % of the variation in biodiversity can be explained by distance, while 95 % is due to other factors or random variation. The positive slope offers minimal support for Hypothesis 1, suggesting that diversity may rise slightly farther into the forest. However, the very low R^2 indicates that the pattern is weak and statistically insignificant, so neither hypothesis is strongly supported. Overall, biodiversity appears relatively stable across the transect, implying that edge effects have little measurable influence on understory diversity in this dataset.

4. Invasive plants such as English holly, Himalayan blackberry, English laurel, and wall lettuce typically share several key life history traits that allow them to spread effectively. They tend to have high reproductive capacity and rapid growth rates, producing many seeds or vegetative shoots that enable them to colonize disturbed areas quickly. In addition, they are tolerant of disturbance and environmental stress, allowing them to thrive in conditions such as high light, trampling, and pollution that many native species cannot withstand. Because of these traits, invasive species are generally expected to be more abundant near the forest edge, where disturbance, sunlight, and human or animal activity are greatest. Deeper within the forest, lower light levels and more stable environmental conditions typically favour shade-tolerant native species, leading to lower invasive abundance.
In Figure 3, the trendline shows a weak positive relationship between distance from the edge and the proportion of total plant cover made up of invasive species ($R^2 = 0.1123$). This means that only about 11% of the variation in invasive cover is explained by distance, indicating a weak and likely non-significant correlation. Although the positive slope slightly contradicts the expectation that invasive abundance should decline farther into the forest, the low R^2 value suggests that the relationship is weak and may result from random variation. Overall, invasive species cover appears highly variable across distances, and the data provide only limited evidence for edge effects. These results are therefore somewhat surprising, as a stronger pattern of higher invasion near the forest edge was expected based on invasive species' life history traits.

5a. Salal is more consistent with a K-selected life strategy. One trait supporting this is that salal is evergreen and long-lived: its leaves live for 2–4 years, and its stems can live many years, producing new growth gradually over time. Another trait is its competitive persistence in forest understory: salal forms dense thickets and often dominates the shrub layer in stable forest environments, tolerating shade and low nutrient soils, rather than rapidly colonizing disturbed open ground. These traits — longevity, shade tolerance, and dominance in stable environments — are more aligned with K-selection than r-selection.

5b. Salal has meaningful importance to Indigenous peoples in the Pacific Northwest. Its berries are eaten fresh or dried into cakes and used in jams and preserves, providing a source of nutrition and traditional food. The leaves are also used medicinally — for example, brewed as teas or applied to treat wounds, burns, and digestive ailments. In local First Nations usage (including Coast Salish groups), salal berries are still collected and used in cultural foods, and the leaves are part of traditional medicines.